WORKED SOLUTION

STEP II 2021, Question 10

Mechanics • Non-Inertial Frames

STEP II 2021, Question 10

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 11.5, *Gravity in an Accelerating Train*: https://beyondstep.keithmcnulty.org/ch11/accelerating-frame.html

The key idea

The axes move with the train, so they are not an inertial frame. The cleanest fix is to work out the bead's *true* acceleration: its acceleration relative to the train plus the train's own acceleration. Newton's second law then holds, and because the wire is smooth the reaction is perpendicular to the wire, so it does no work along the bead's path. Dotting the equation of motion with the bead's (relative) velocity removes the unknown reaction and leaves an energy-like quantity that is conserved.

Solution

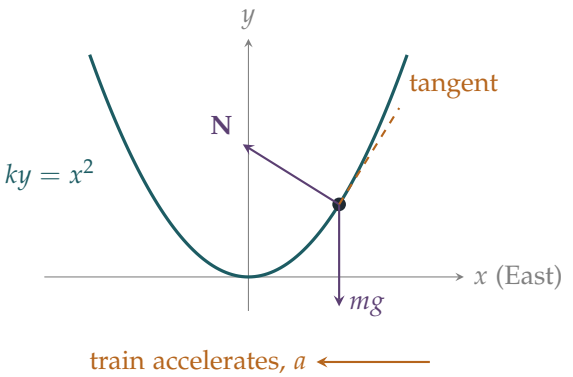
Model the bead as a particle of mass m and the wire as smooth, so the reaction $\mathbf{N}$ of the wire on the bead is always perpendicular to the wire. Write $\mathbf{i}$ and $\mathbf{j}$ for unit vectors east and up.

The train accelerates westwards, so its acceleration is $-a\mathbf{i}$. The bead's position relative to the train is $x\mathbf{i} + y\mathbf{j}$, so its true (ground-frame) acceleration is

$$(\ddot{x} - a)\,\mathbf{i} + \ddot{y}\,\mathbf{j}.$$

The only forces on the bead are its weight $-mg\mathbf{j}$ and $\mathbf{N}$, so Newton's second law gives

$$m(\ddot{x} - a)\,\mathbf{i} + m\ddot{y}\,\mathbf{j} = \mathbf{N} - mg\,\mathbf{j}. \qquad (*)$$



Part (i)

Suppose the bead stayed at the origin relative to the train. Then it would share the train's acceleration, $-a\mathbf{i}$, so it would need a horizontal force $-ma\mathbf{i}$ with $a > 0$. But at the origin the wire $y = x^2/k$ has gradient $2x/k = 0$, so it is horizontal there, and the reaction from a smooth wire is perpendicular to it, which means vertical. Gravity is vertical too. No horizontal force acts, so the bead cannot accelerate with the train; it must move relative to the train. (Its true horizontal acceleration is 0 while the train's is a westwards, so the bead starts to slide east along the wire, towards positive x.)

Part (ii)

While the bead moves along the wire, its velocity relative to the train, $\dot{x}\,\mathbf{i} + \dot{y}\,\mathbf{j}$, is tangent to the wire, because the wire is fixed in the train. Since $\mathbf{N}$ is perpendicular to the wire,

$$\mathbf{N} \cdot (\dot{x}\,\mathbf{i} + \dot{y}\,\mathbf{j}) = 0.$$

Rearranging $(*)$ as $m(\ddot{x} - a)\,\mathbf{i} + m(\ddot{y} + g)\,\mathbf{j} = \mathbf{N}$ and taking the scalar product of both sides with $\dot{x}\,\mathbf{i} + \dot{y}\,\mathbf{j}$ gives, after dividing by m,

$$\dot{x}(\ddot{x} - a) + \dot{y}(\ddot{y} + g) = 0,$$

as required. The left-hand side is an exact derivative:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\tfrac{1}{2}(\dot{x}^2 + \dot{y}^2) - ax + gy\right) = \dot{x}\ddot{x} + \dot{y}\ddot{y} - a\dot{x} + g\dot{y} = 0,$$

so $\frac{1}{2}(\dot{x}^2 + \dot{y}^2) - ax + gy$ is constant during the motion.

Ask yourself Why is it legitimate to dot with the *relative* velocity, when the reaction does work in the ground frame?

In the ground frame the bead's velocity has an extra $-at\,\mathbf{i}$, which is not along the wire, so $\mathbf{N}$ does work there. But the identity $\mathbf{N} \cdot \mathbf{u} = 0$ is pure geometry: it holds for any vector $\mathbf{u}$ tangent to the wire, and the relative velocity is such a vector. That is all we used.

Part (iii)

Initially $x = y = 0$ and $\dot{x} = \dot{y} = 0$, so the constant is 0. Writing v for the speed relative to the train and using $y = x^2/k$,

$$\tfrac{1}{2}v^2 = ax - gy = ax - \frac{gx^2}{k} = \frac{g}{k}\,x\Big(\frac{ak}{g} - x\Big). \qquad (\dagger)$$

Since $v^2 \geq 0$, the bead is confined to $0 \leq x \leq ak/g$, and $v = 0$ only at the two ends of this interval. By part (i) the bead starts moving towards positive x. While $0 < x < ak/g$ its speed is positive, so it keeps moving along the wire in the same direction until it reaches $x = ak/g$, where it momentarily stops. (It does turn back there: the tangential component of $(*)$ at that point is not zero, since $\frac{\mathrm{d}}{\mathrm{d}x}(ax - gx^2/k) = a - 2a = -a \neq 0$.) The bead then oscillates between $x = 0$ and $x = ak/g$.

On this interval $y = x^2/k$ is increasing in x, so the greatest height is reached at $x = ak/g$:

$$b = \frac{1}{k}\Big(\frac{ak}{g}\Big)^2 = \frac{a^2k}{g^2}.$$

Part (iv)

By $(\dagger)$, $v^2 = 2\big(ax - gx^2/k\big)$, a quadratic in x with a maximum at the midpoint of its roots 0 and ak/g, so the speed relative to the train is greatest when

$$x = \frac{ak}{2g}.$$

There

$$v^2 = 2\Big(\frac{a^2k}{2g} - \frac{a^2k}{4g}\Big) = \frac{a^2k}{2g}, \qquad \text{so} \qquad v_{\max} = a\sqrt{\frac{k}{2g}}.$$

Checks

- Limiting case: if $a = 0$ the bead simply stays at the bottom of the wire, and indeed $b = 0$ and $v_{\max} = 0$.
- Effective gravity: in the train the bead feels an effective gravity $(a, -g)$, tilted towards the east. The point of fastest motion, $x = ak/(2g)$, is exactly where the wire's gradient $2x/k$ equals a/g, so the tangent is perpendicular to the effective gravity: the "lowest point" of the wire in the tilted field. The turning point $x = ak/g$ is twice as far out, as for a symmetric oscillation in a potential that is quadratic in x.
- Dimensions: k is a length, so a^2k/g^2 is a length and $a\sqrt{k/g}$ is a speed.
- Numerically, with $a = 2$, $k = 1.5$ and $g = 9.8$ (SI units), stepping the motion forward by computer gives a greatest height of $0.06247\,\mathrm{m}$ and a greatest speed of $0.5533\,\mathrm{m\,s^{-1}}$ at $x = 0.1531\,\mathrm{m}$, matching a^2k/g^2, $a\sqrt{k/(2g)}$ and $ak/(2g)$.

Common slips

- Writing $\ddot{x}$ for the bead's true horizontal acceleration. That is its acceleration relative to the train; the train's own acceleration $-a$ must be added.
- Getting the sign of a wrong: the train accelerates *west*, so in the train the bead is pushed *east*, towards positive x.
- Taking the speed to be greatest at the bottom of the wire, $x = 0$. In the train the effective "down" is tilted, so the fastest point is at $x = ak/(2g)$ instead.

Beyond STEP

Part (ii) is a first taste of mechanics in non-inertial frames: in the train's frame the term $-ax$ acts like a potential energy for a uniform "fictitious" force ma eastwards, and adding it to gravity gives a tilted effective gravity. At university this extends to rotating frames, where centrifugal and Coriolis forces appear, and to Lagrangian mechanics, where constraint forces like $\mathbf{N}$ are eliminated from the start. Kleppner and Kolenkow, *An Introduction to Mechanics*, has a good chapter on non-inertial systems and fictitious forces.

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